西 南 交 通 大 学

《数据库与云计算原理》课程作业

设计报告

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第一章 结构设计

1.1 概念结构设计

1.1.1 需求和功能

需求：

1. 管理工厂信息：厂名和厂长名。
2. 管理车间信息：车间号、车间主任姓名、地址和电话。
3. 管理工人信息：职工号、姓名、出生日期、性别和工种。
4. 管理产品信息：产品号和价格。
5. 管理零件信息：零件号、重量和价格。
6. 管理产品与零件的组成关系，以及零件在多个车间的生产情况。
7. 管理仓库信息：仓库号、仓库主管姓名和电话。
8. 管理产品和零件的库存信息。

功能：

1. 添加、删除、更新和查询工厂、车间、工人、产品和零件信息。
2. 管理产品与零件的组装关系。
3. 管理仓库信息，包括库存的添加、删除、更新和查询。

1.1.2 实体及属性

1. 工厂(Factory)

厂名(FactoryName)：唯一标识工厂的名称。

厂长名(ManagerName)：工厂负责人的姓名。

2. 车间(Workshop)

车间号(WorkshopID)：唯一标识车间的编号。

车间主任姓名(DirectorName)：车间负责人的姓名。

地址(Address)：车间的地理位置。

电话(Phone)：车间的联系电话。

3. 工人(Worker)

职工号(EmployeeID)：唯一标识工人的编号。

姓名(Name)：工人的姓名。

出生日期(BirthDate)：工人的出生日期。

性别(Gender)：工人的性别。

工种(Position)：工人的工作岗位。

车间号(WorkshopID)：工人所属车间的编号。

4. 产品(Product)

产品号(ProductID)：唯一标识产品的编号。

价格(Price)：产品的价格。

5. 零件(Part)

零件号(PartID)：唯一标识零件的编号。

重量(Weight)：零件的重量。

价格(Price)：零件的价格。

6. 仓库(Warehouse)

仓库号(WarehouseID)：唯一标识仓库的编号。

仓库主管姓名(ManagerName)：仓库负责人的姓名。

电话(Phone)：仓库的联系电话。

1.1.3 实体联系及联系属性

关系	联系	联系属性
工厂-车间 (Factory-Workshop)	一个工厂可以有多个车间，一个车间属于一个工厂。	无额外联系属性，通过工厂名(FactoryName)关联。
车间-工人 (Workshop-Worker)	一个车间可以有多个工人，一个工人只能在一个车间工作。	车间号(WorkshopID)作为外键在工人表中。
车间-产品 (Workshop-Product)	一个车间可以生产多种产品，一个产品可以由一个或多个车间生产。	无额外联系属性，通过车间号(WorkshopID)关联。
车间-零件 (Workshop-Part)	一个车间可以制造多种零件，一个零件可以由一个或多个车间制造。	无额外联系属性，通过车间号(WorkshopID)关联。
产品-零件 (Product-Part)	一个产品由多种零件组成，一种零件可以用于多种产品的装配。	需要一个关联表来表示这种多对多关系，包含产品号(ProductID)和零件号(PartID)。
仓库-产品/零件 (Warehouse-Product/Part	产品和零件均存入仓库中，一个仓库可以存储多种产品和零件。	需要两个关联表来表示这种多对多关系，分别是仓库产品关联表(WarehouseProduct)和仓库零件关联表(WarehousePart)，都包含仓库号(WarehouseID)和产品号/零件号(ProductID/PartID)。

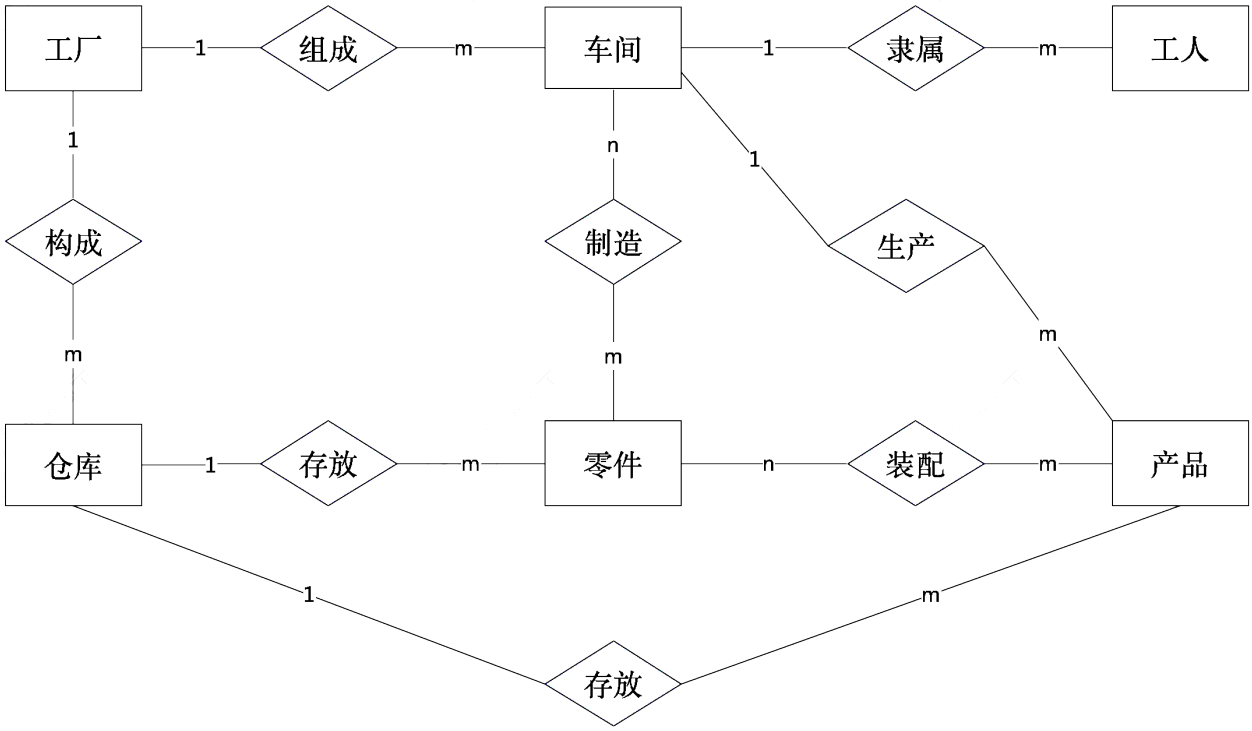
1.1.4 数据字典

```
ENTITY Factory {
    FactoryName VARCHAR(255) PRIMARY KEY,
    ManagerName VARCHAR(255)
}

ENTITY Workshop {
    WorkshopID INT PRIMARY KEY AUTO_INCREMENT,
    DirectorName VARCHAR(255),
    Address VARCHAR(255),
    Phone VARCHAR(11),
    FactoryName VARCHAR(255) FOREIGN KEY REFERENCES
Factory(FactoryName)
}

ENTITY Worker {
    EmployeeID INT PRIMARY KEY AUTO_INCREMENT,
    Name VARCHAR(255),
    BirthDate DATE,
    Gender ENUM('M', 'F'),
    Position VARCHAR(255),
    WorkshopID INT FOREIGN KEY REFERENCES Workshop(WorkshopID)
}

ENTITY Product {
    ProductID INT PRIMARY KEY AUTO_INCREMENT,
    Price DECIMAL(10, 2)
}
```

```
ENTITY Part {
    PartID INT PRIMARY KEY AUTO_INCREMENT,
    Weight DECIMAL(10, 2),
    Price DECIMAL(10, 2)
}

ENTITY Warehouse {
    WarehouseID INT PRIMARY KEY AUTO_INCREMENT,
    ManagerName VARCHAR(255),
    Phone VARCHAR(11)
}

ENTITY ProductPart {
    ProductID INT,
    PartID INT,
    PRIMARY KEY (ProductID, PartID),
    FOREIGN KEY (ProductID) REFERENCES Product(ProductID),
    FOREIGN KEY (PartID) REFERENCES Part(PartID)
}

ENTITY WarehouseProduct {
    WarehouseID INT,
    ProductID INT,
    PRIMARY KEY (WarehouseID, ProductID),
    FOREIGN KEY (WarehouseID) REFERENCES Warehouse(WarehouseID),
    FOREIGN KEY (ProductID) REFERENCES Product(ProductID)
}

ENTITY WarehousePart {
    WarehouseID INT,
```

```
    PartID INT,
    PRIMARY KEY (WarehouseID, PartID),
    FOREIGN KEY (WarehouseID) REFERENCES Warehouse(WarehouseID),
    FOREIGN KEY (PartID) REFERENCES Part(PartID)
}
```

1.2 E-R 图



ER 图展示了实体关系模型，包括工厂、车间、工人、产品、零件和仓库六个实体。工厂由多个车间组成，每个车间有多个工人，且工人只隶属于一个车间。车间负责生产多种产品和零件，产品由多种零件装配而成，而零件可以被用于多种产品的生产。产品和零件都可以存放在仓库中，一个仓库可以存放多种产品和零件。

图中的 1, n, m 表示实体间的联系数量，例如，一个工厂（1）可以有多个车间（n），而一个车间（1）可以有多个工人（n）。

其中产品与零件、零件与车间之间的关系是多对多（m:n），需要中间表。

1.3 逻辑结构

1.3.1 关系模型及数据表格

关系（表）	字段	描述
Factory	FactoryName	VARCHAR(255), Primary Key
	ManagerName	VARCHAR(255)
Workshop	WorkshopID	INT, Primary Key, Auto Increment
	DirectorName	VARCHAR(255)
	Address	VARCHAR(255)
	Phone	VARCHAR(11)
	FactoryName	VARCHAR(255), Foreign Key References Factory(FactoryName)
Worker	EmployeeID	INT, Primary Key, Auto Increment
	Name	VARCHAR(255)
	BirthDate	DATE
	Gender	ENUM('M', 'F')
	Position	VARCHAR(255)
	WorkshopID	INT, Foreign Key References Workshop(WorkshopID)
Product	ProductID	INT, Primary Key, Auto Increment
	Price	DECIMAL(10, 2)
Part	PartID	INT, Primary Key, Auto Increment
	Weigh	DECIMAL(10, 2)
	Price	DECIMAL(10, 2)

Warehouse	WarehouseID	INT, Primary Key, Auto Increment
	ManagerName	VARCHAR(255)
	Phone	VARCHAR(11)
ProductPart	ProductID	INT, Primary Key, Foreign Key References
	PartID	INT, Primary Key, Foreign Key References Part(PartID)
WarehouseProduct	WarehouseID	INT, Primary Key, Foreign Key References Warehouse(WarehouseID)
	ProductID	INT, Primary Key, Foreign Key References Product(ProductID)
WarehousePart	WarehouseID	INT, Primary Key, Foreign Key References Warehouse(WarehouseID)
	PartID	INT, Primary Key, Foreign Key References Part(PartID)

第二章 SQL 语句

2.1 创建表、视图、索引

2.1.1 创建表

```
CREATE TABLE Factory (
    FactoryName VARCHAR(255) PRIMARY KEY,
    ManagerName VARCHAR(255)
);
```

```
CREATE TABLE Workshop (
    WorkshopID INT AUTO_INCREMENT PRIMARY KEY,
    DirectorName VARCHAR(255),
    Address VARCHAR(255),
    Phone VARCHAR(11),
    FactoryName VARCHAR(255),
    FOREIGN KEY (FactoryName) REFERENCES Factory(FactoryName)
);

CREATE TABLE Worker (
    EmployeeID INT AUTO_INCREMENT PRIMARY KEY,
    Name VARCHAR(255),
    BirthDate DATE,
    Gender ENUM('M', 'F'),
    Position VARCHAR(255),
    WorkshopID INT,
    FOREIGN KEY (WorkshopID) REFERENCES Workshop(WorkshopID)
);

CREATE TABLE Product (
    ProductID INT AUTO_INCREMENT PRIMARY KEY,
    Price DECIMAL(10, 2)
);

CREATE TABLE Part (
    PartID INT AUTO_INCREMENT PRIMARY KEY,
    Weight DECIMAL(10, 2),
    Price DECIMAL(10, 2)
);
```

```
CREATE TABLE Warehouse (
    WarehouseID INT AUTO_INCREMENT PRIMARY KEY,
    ManagerName VARCHAR(255),
    Phone VARCHAR(11)
);

CREATE TABLE ProductPart (
    ProductID INT,
    PartID INT,
    PRIMARY KEY (ProductID, PartID),
    FOREIGN KEY (ProductID) REFERENCES Product(ProductID),
    FOREIGN KEY (PartID) REFERENCES Part(PartID)
);

CREATE TABLE WarehouseProduct (
    WarehouseID INT,
    ProductID INT,
    PRIMARY KEY (WarehouseID, ProductID),
    FOREIGN KEY (WarehouseID) REFERENCES Warehouse(WarehouseID),
    FOREIGN KEY (ProductID) REFERENCES Product(ProductID)
);

CREATE TABLE WarehousePart (
    WarehouseID INT,
    PartID INT,
    PRIMARY KEY (WarehouseID, PartID),
    FOREIGN KEY (WarehouseID) REFERENCES Warehouse(WarehouseID),
    FOREIGN KEY (PartID) REFERENCES Part(PartID)
);
```

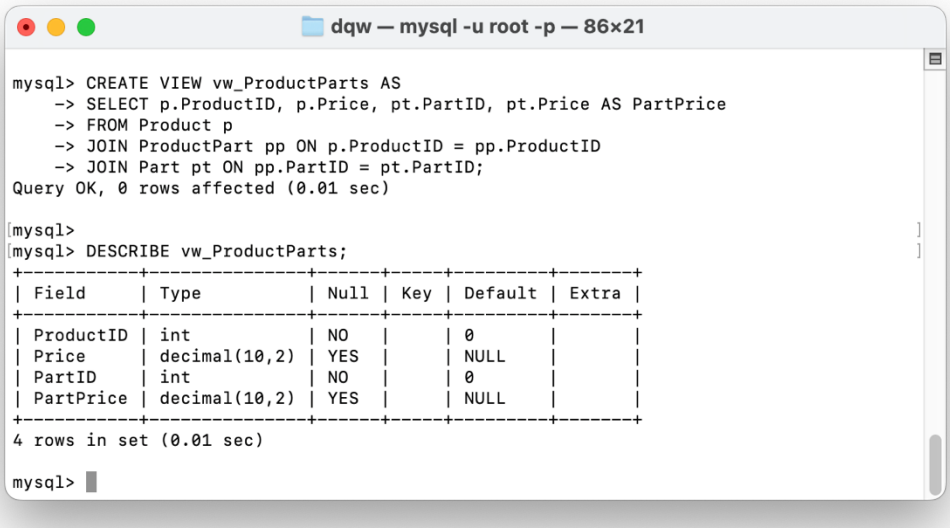
dqw — mysql -u root -p — 86×35

```
    ->     WarehouseID INT,
    ->     ProductID INT,
    ->     PRIMARY KEY (WarehouseID, ProductID),
    ->     FOREIGN KEY (WarehouseID) REFERENCES Warehouse(WarehouseID),
    ->     FOREIGN KEY (ProductID) REFERENCES Product(ProductID)
    -> );
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> CREATE TABLE WarehousePart (
    ->     WarehouseID INT,
    ->     PartID INT,
    ->     PRIMARY KEY (WarehouseID, PartID),
    ->     FOREIGN KEY (WarehouseID) REFERENCES Warehouse(WarehouseID),
    ->     FOREIGN KEY (PartID) REFERENCES Part(PartID)
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> SHOW TABLES;
+------------------+
| Tables_in_mydb   |
+------------------+
| Factory          |
| Part             |
| Product          |
| ProductPart      |
| Warehouse        |
| WarehousePart    |
| WarehouseProduct |
| Worker           |
| Workshop         |
+------------------+
9 rows in set (0.00 sec)

mysql>
```

2.1.2 创建视图

以产品及其组成零件为例，将产品 ID、产品价格、零件 ID 和零件通过 ProductPart 关联表将 Product 和 Part 表连接起来。

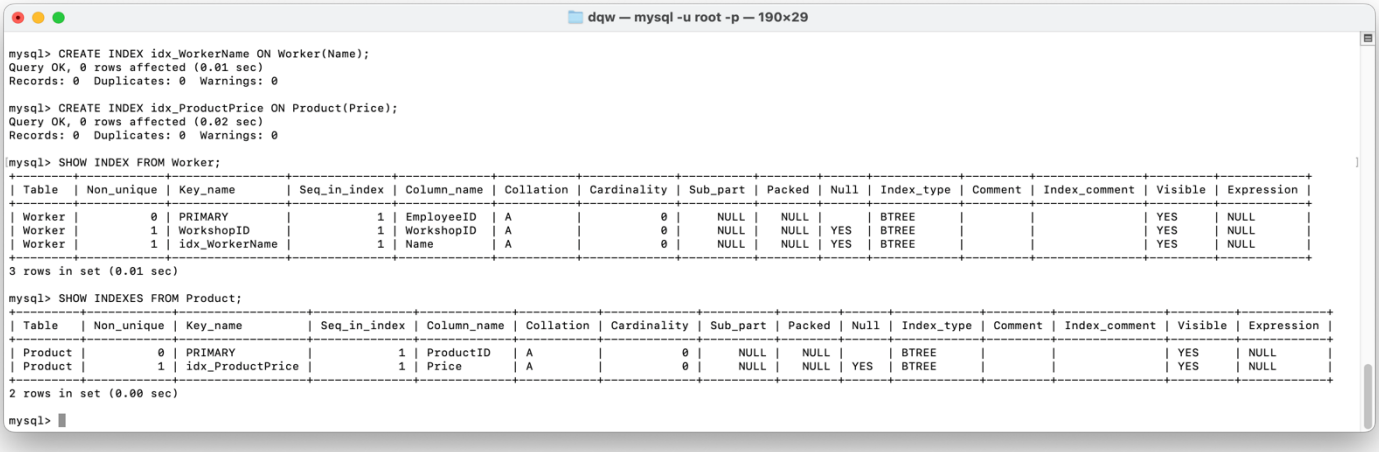
```
CREATE VIEW vw_ProductParts AS
SELECT p.ProductID, p.Price, pt.PartID, pt.Price AS PartPrice
FROM Product p
JOIN ProductPart pp ON p.ProductID = pp.ProductID
JOIN Part pt ON pp.PartID = pt.PartID;
```



2.1.3 创建索引

以基于工人姓名、产品价格查询为例。

```
CREATE INDEX idx_WorkerName ON Worker(Name);
CREATE INDEX idx_ProductPrice ON Product(Price);
```



```
dqw — mysql -u root -p — 190×29

mysql> CREATE INDEX idx_WorkerName ON Worker(Name);
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> CREATE INDEX idx_ProductPrice ON Product(Price);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> SHOW INDEX FROM Worker;
+--------+------------+-------------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| Table  | Non_unique | Key_name          | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expression |
+--------+------------+-------------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| Worker |          0 | PRIMARY           |            1 | EmployeeID  | A         |           0 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| Worker |          1 | WorkshopID        |            1 | WorkshopID  | A         |           0 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL       |
| Worker |          1 | idx_WorkerName    |            1 | Name        | A         |           0 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL       |
+--------+------------+-------------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
3 rows in set (0.01 sec)

mysql> SHOW INDEXES FROM Product;
+---------+------------+-------------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| Table   | Non_unique | Key_name          | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expression |
+---------+------------+-------------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| Product |          0 | PRIMARY           |            1 | ProductID   | A         |           0 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| Product |          1 | idx_ProductPrice  |            1 | Price       | A         |           0 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL       |
+---------+------------+-------------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
2 rows in set (0.00 sec)

mysql>
```

2.2 插入测试数据

INSERT INTO Factory (FactoryName, ManagerName) VALUES ('工厂 A', '张三');

INSERT INTO Workshop (WorkshopID, DirectorName, Address, Phone, FactoryName) VALUES

(1, '李四', '地址 1', '12345678901', '工厂 A'),

(2, '王五', '地址 2', '12345678902', '工厂 A');

INSERT INTO Worker (Name, BirthDate, Gender, Position, WorkshopID) VALUES

('工人 1', '1990-01-01', 'M', '工程师', 1),

('工人 2', '1991-02-02', 'F', '技术员', 1),

('工人 3', '1992-03-03', 'M', '工程师', 1),

('工人 4', '1993-04-04', 'F', '工程师', 2),

('工人 5', '1994-05-05', 'M', '技术员', 2),

('工人 6', '1995-06-06', 'F', '工程师', 2);

INSERT INTO Part (PartID, Weight, Price) VALUES

(1, 10.5, 100.0), -- 零件 A

(2, 20.5, 200.0), -- 零件 a

```
(3, 15.5, 150.0), --  零件 B
(4, 25.5, 250.0); --  零件 b

INSERT INTO Product (ProductID, Price) VALUES
(1, 500.0),
(2, 600.0),
(3, 700.0);

INSERT INTO ProductPart (ProductID, PartID) VALUES
(1, 1),
(1, 2),
(2, 1),
(2, 3),
(2, 4),
(3, 2),
(3, 3);

INSERT INTO Warehouse (WarehouseID, ManagerName, Phone) VALUES
(1, '赵六', '12345678903'),
(2, '孙七', '12345678904');

INSERT INTO WarehouseProduct (WarehouseID, ProductID) VALUES
(1, 1),
(2, 2);

INSERT INTO WarehousePart (WarehouseID, PartID) VALUES
(1, 1),
(2, 3);
```

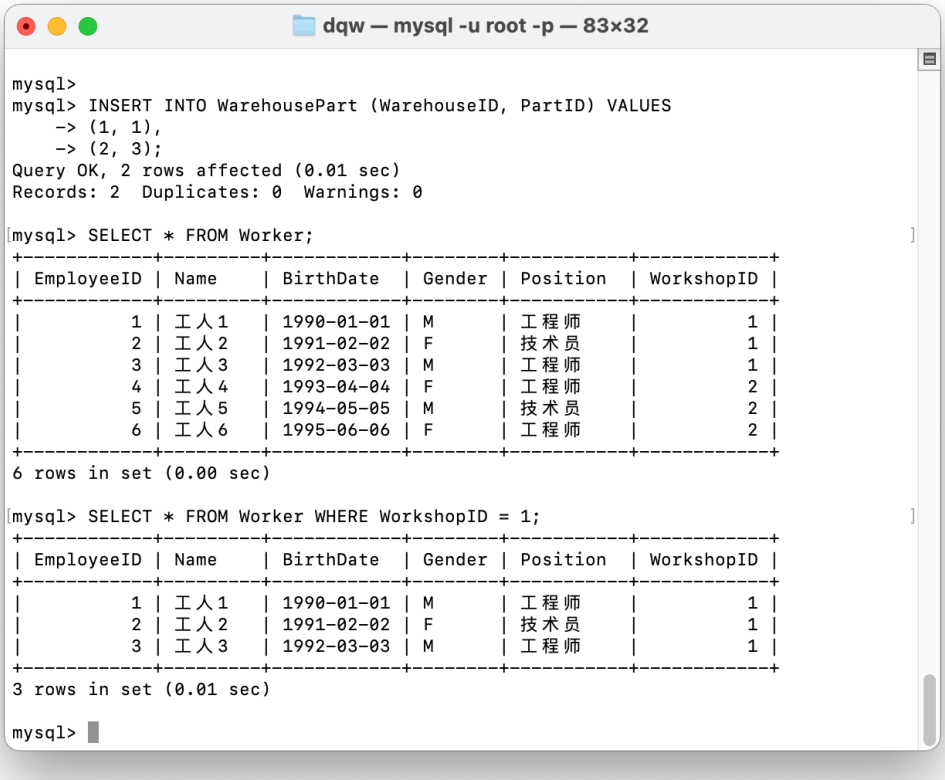
2.3 基本查询

#查询所有工人信息

SELECT * FROM Worker;

#查询特定车间的工人

SELECT * FROM Worker WHERE WorkshopID = 1;



```
dqw — mysql -u root -p — 83×32

mysql>
mysql> INSERT INTO WarehousePart (WarehouseID, PartID) VALUES
    -> (1, 1),
    -> (2, 3);
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Worker;
+------------+---------+------------+--------+-----------+------------+
| EmployeeID | Name    | BirthDate  | Gender | Position  | WorkshopID |
+------------+---------+------------+--------+-----------+------------+
|          1 | 工人1   | 1990-01-01 | M      | 工程师    |          1 |
|          2 | 工人2   | 1991-02-02 | F      | 技术员    |          1 |
|          3 | 工人3   | 1992-03-03 | M      | 工程师    |          1 |
|          4 | 工人4   | 1993-04-04 | F      | 工程师    |          2 |
|          5 | 工人5   | 1994-05-05 | M      | 技术员    |          2 |
|          6 | 工人6   | 1995-06-06 | F      | 工程师    |          2 |
+------------+---------+------------+--------+-----------+------------+
6 rows in set (0.00 sec)

mysql> SELECT * FROM Worker WHERE WorkshopID = 1;
+------------+---------+------------+--------+-----------+------------+
| EmployeeID | Name    | BirthDate  | Gender | Position  | WorkshopID |
+------------+---------+------------+--------+-----------+------------+
|          1 | 工人1   | 1990-01-01 | M      | 工程师    |          1 |
|          2 | 工人2   | 1991-02-02 | F      | 技术员    |          1 |
|          3 | 工人3   | 1992-03-03 | M      | 工程师    |          1 |
+------------+---------+------------+--------+-----------+------------+
3 rows in set (0.01 sec)

mysql>
```

#查询包含特定零件的产品

SELECT p.ProductID, p.Price, GROUP_CONCAT(pt.PartID) AS PartIDs

FROM Product p

JOIN ProductPart pp ON p.ProductID = pp.ProductID

JOIN Part pt ON pp.PartID = pt.PartID

WHERE pt.PartID = 1 OR pt.PartID = 3

GROUP BY p.ProductID;

```
mysql> SELECT p.ProductID, p.Price, GROUP_CONCAT(pt.PartID) AS PartIDs
    -> FROM Product p
    -> JOIN ProductPart pp ON p.ProductID = pp.ProductID
    -> JOIN Part pt ON pp.PartID = pt.PartID
    -> WHERE pt.PartID = 1 OR pt.PartID = 3
    -> GROUP BY p.ProductID;
+-----------+--------+---------+
| ProductID | Price  | PartIDs |
+-----------+--------+---------+
|         1 | 500.00 | 1       |
|         2 | 600.00 | 1,3     |
|         3 | 700.00 | 3       |
+-----------+--------+---------+
3 rows in set (0.00 sec)

mysql>
```

2.4 更改

#更新工人信息

UPDATE Worker SET Position = '高级工程师' WHERE EmployeeID = 1;

#更新产品价格

UPDATE Product SET Price = 520.0 WHERE ProductID = 1;

```
mysql> UPDATE Worker SET Position = '高级工程师' WHERE EmployeeID = 1;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE Product SET Price = 520.0 WHERE ProductID = 1;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM Worker WHERE EmployeeID = 1;
+------------+---------+------------+--------+-----------------+------------+
| EmployeeID | Name    | BirthDate  | Gender | Position        | WorkshopID |
+------------+---------+------------+--------+-----------------+------------+
|          1 | 工人1   | 1990-01-01 | M      | 高级工程师      |          1 |
+------------+---------+------------+--------+-----------------+------------+
1 row in set (0.00 sec)

mysql> SELECT * FROM Product WHERE ProductID = 1;
+-----------+--------+
| ProductID | Price  |
+-----------+--------+
|         1 | 520.00 |
+-----------+--------+
1 row in set (0.00 sec)

mysql>
```

2.5 删除

#删除工人

DELETE FROM Worker WHERE EmployeeID = 6;

2.6 结果

-- MySQL dump 10.13 Distrib 5.7.24, for osx11.1 (x86_64)

--

-- Host: localhost Database: MyDb

-- --

-- Server version 8.0.39

/*!40101 SET @OLD_CHARACTER_SET_CLIENT=@@CHARACTER_SET_CLIENT */;

/*!40101 SET @OLD_CHARACTER_SET_RESULTS=@@CHARACTER_SET_RESULTS */;

/*!40101 SET @OLD_COLLATION_CONNECTION=@@COLLATION_CONNECTION */;

/*!40101 SET NAMES utf8 */;

/*!40103 SET @OLD_TIME_ZONE=@@TIME_ZONE */;

/*!40103 SET TIME_ZONE='+00:00' */;

/*!40014 SET @OLD_UNIQUE_CHECKS=@@UNIQUE_CHECKS, UNIQUE_CHECKS=0 */;

/*!40014 SET @OLD_FOREIGN_KEY_CHECKS=@@FOREIGN_KEY_CHECKS, FOREIGN_KEY_CHECKS=0 */;

/*!40101 SET @OLD_SQL_MODE=@@SQL_MODE, SQL_MODE='NO_AUTO_VALUE_ON_ZERO' */;

/*!40111 SET @OLD_SQL_NOTES=@@SQL_NOTES, SQL_NOTES=0 */;

```
--
-- Table structure for table `Factory`
--

DROP TABLE IF EXISTS `Factory`;
/*!40101 SET @saved_cs_client     = @@character_set_client */;
/*!40101 SET character_set_client = utf8 */;
CREATE TABLE `Factory` (
  `FactoryName` varchar(255) NOT NULL,
  `ManagerName` varchar(255) DEFAULT NULL,
  PRIMARY KEY (`FactoryName`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4
COLLATE=utf8mb4_0900_ai_ci;
/*!40101 SET character_set_client = @saved_cs_client */;

--
-- Dumping data for table `Factory`
--

LOCK TABLES `Factory` WRITE;
/*!40000 ALTER TABLE `Factory` DISABLE KEYS */;
INSERT INTO `Factory` VALUES ('工厂 A','张三');
/*!40000 ALTER TABLE `Factory` ENABLE KEYS */;
UNLOCK TABLES;

--
-- Table structure for table `Part`
--
```

```
DROP TABLE IF EXISTS `Part`;
/*!40101 SET @saved_cs_client     = @@character_set_client */;
/*!40101 SET character_set_client = utf8 */;
CREATE TABLE `Part` (
  `PartID` int NOT NULL AUTO_INCREMENT,
  `Weight` decimal(10,2) DEFAULT NULL,
  `Price` decimal(10,2) DEFAULT NULL,
  PRIMARY KEY (`PartID`)
) ENGINE=InnoDB AUTO_INCREMENT=5 DEFAULT CHARSET=utf8mb4
COLLATE=utf8mb4_0900_ai_ci;
/*!40101 SET character_set_client = @saved_cs_client */;

--
-- Dumping data for table `Part`
--

LOCK TABLES `Part` WRITE;
/*!40000 ALTER TABLE `Part` DISABLE KEYS */;
INSERT INTO `Part` VALUES
(1,10.50,100.00),(2,20.50,200.00),(3,15.50,150.00),(4,25.50,250.00);
/*!40000 ALTER TABLE `Part` ENABLE KEYS */;
UNLOCK TABLES;

--
-- Table structure for table `Product`
--

DROP TABLE IF EXISTS `Product`;
/*!40101 SET @saved_cs_client     = @@character_set_client */;
/*!40101 SET character_set_client = utf8 */;
```

```
CREATE TABLE `Product` (
  `ProductID` int NOT NULL AUTO_INCREMENT,
  `Price` decimal(10,2) DEFAULT NULL,
  PRIMARY KEY (`ProductID`),
  KEY `idx_ProductPrice` (`Price`)
) ENGINE=InnoDB AUTO_INCREMENT=4 DEFAULT CHARSET=utf8mb4
COLLATE=utf8mb4_0900_ai_ci;
/*!40101 SET character_set_client = @saved_cs_client */;

--
-- Dumping data for table `Product`
--

LOCK TABLES `Product` WRITE;
/*!40000 ALTER TABLE `Product` DISABLE KEYS */;
INSERT INTO `Product` VALUES (1,520.00),(2,600.00),(3,700.00);
/*!40000 ALTER TABLE `Product` ENABLE KEYS */;
UNLOCK TABLES;

--
-- Table structure for table `ProductPart`
--

DROP TABLE IF EXISTS `ProductPart`;
/*!40101 SET @saved_cs_client     = @@character_set_client */;
/*!40101 SET character_set_client = utf8 */;
CREATE TABLE `ProductPart` (
  `ProductID` int NOT NULL,
  `PartID` int NOT NULL,
  PRIMARY KEY (`ProductID`,`PartID`),
```

```
  KEY `PartID` (`PartID`),
  CONSTRAINT `productpart_ibfk_1` FOREIGN KEY (`ProductID`)
REFERENCES `Product` (`ProductID`),
  CONSTRAINT `productpart_ibfk_2` FOREIGN KEY (`PartID`) REFERENCES
`Part` (`PartID`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4
COLLATE=utf8mb4_0900_ai_ci;
/*!40101 SET character_set_client = @saved_cs_client */;

--
-- Dumping data for table `ProductPart`
--

LOCK TABLES `ProductPart` WRITE;
/*!40000 ALTER TABLE `ProductPart` DISABLE KEYS */;
INSERT INTO `ProductPart` VALUES (1,1),(2,1),(1,2),(3,2),(2,3),(3,3),(2,4);
/*!40000 ALTER TABLE `ProductPart` ENABLE KEYS */;
UNLOCK TABLES;

--
-- Temporary table structure for view `vw_productparts`
--

DROP TABLE IF EXISTS `vw_productparts`;
/*!50001 DROP VIEW IF EXISTS `vw_productparts`*/;
SET @saved_cs_client     = @@character_set_client;
SET character_set_client = utf8;
/*!50001 CREATE VIEW `vw_productparts` AS SELECT
 1 AS `ProductID`,
 1 AS `Price`,
```

```
  1 AS `PartID`,
  1 AS `PartPrice`*/;
SET character_set_client = @saved_cs_client;

--
-- Table structure for table `Warehouse`
--

DROP TABLE IF EXISTS `Warehouse`;
/*!40101 SET @saved_cs_client     = @@character_set_client */;
/*!40101 SET character_set_client = utf8 */;
CREATE TABLE `Warehouse` (
  `WarehouseID` int NOT NULL AUTO_INCREMENT,
  `ManagerName` varchar(255) DEFAULT NULL,
  `Phone` varchar(11) DEFAULT NULL,
  PRIMARY KEY (`WarehouseID`)
) ENGINE=InnoDB AUTO_INCREMENT=3 DEFAULT CHARSET=utf8mb4
COLLATE=utf8mb4_0900_ai_ci;
/*!40101 SET character_set_client = @saved_cs_client */;

--
-- Dumping data for table `Warehouse`
--

LOCK TABLES `Warehouse` WRITE;
/*!40000 ALTER TABLE `Warehouse` DISABLE KEYS */;
INSERT INTO `Warehouse` VALUES (1,'赵六','12345678903'),(2,'孙七
','12345678904');
/*!40000 ALTER TABLE `Warehouse` ENABLE KEYS */;
UNLOCK TABLES;
```

```
--
-- Table structure for table `WarehousePart`
--

DROP TABLE IF EXISTS `WarehousePart`;
/*!40101 SET @saved_cs_client     = @@character_set_client */;
/*!40101 SET character_set_client = utf8 */;
CREATE TABLE `WarehousePart` (
  `WarehouseID` int NOT NULL,
  `PartID` int NOT NULL,
  PRIMARY KEY (`WarehouseID`,`PartID`),
  KEY `PartID` (`PartID`),
  CONSTRAINT `warehousepart_ibfk_1` FOREIGN KEY (`WarehouseID`)
REFERENCES `Warehouse` (`WarehouseID`),
  CONSTRAINT `warehousepart_ibfk_2` FOREIGN KEY (`PartID`)
REFERENCES `Part` (`PartID`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4
COLLATE=utf8mb4_0900_ai_ci;
/*!40101 SET character_set_client = @saved_cs_client */;

--
-- Dumping data for table `WarehousePart`
--

LOCK TABLES `WarehousePart` WRITE;
/*!40000 ALTER TABLE `WarehousePart` DISABLE KEYS */;
INSERT INTO `WarehousePart` VALUES (1,1),(2,3);
/*!40000 ALTER TABLE `WarehousePart` ENABLE KEYS */;
UNLOCK TABLES;
```

```
--
-- Table structure for table `WarehouseProduct`
--

DROP TABLE IF EXISTS `WarehouseProduct`;
/*!40101 SET @saved_cs_client     = @@character_set_client */;
/*!40101 SET character_set_client = utf8 */;
CREATE TABLE `WarehouseProduct` (
  `WarehouseID` int NOT NULL,
  `ProductID` int NOT NULL,
  PRIMARY KEY (`WarehouseID`,`ProductID`),
  KEY `ProductID` (`ProductID`),
  CONSTRAINT `warehouseproduct_ibfk_1` FOREIGN KEY (`WarehouseID`)
REFERENCES `Warehouse` (`WarehouseID`),
  CONSTRAINT `warehouseproduct_ibfk_2` FOREIGN KEY (`ProductID`)
REFERENCES `Product` (`ProductID`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4
COLLATE=utf8mb4_0900_ai_ci;
/*!40101 SET character_set_client = @saved_cs_client */;

--
-- Dumping data for table `WarehouseProduct`
--

LOCK TABLES `WarehouseProduct` WRITE;
/*!40000 ALTER TABLE `WarehouseProduct` DISABLE KEYS */;
INSERT INTO `WarehouseProduct` VALUES (1,1),(2,2);
/*!40000 ALTER TABLE `WarehouseProduct` ENABLE KEYS */;
UNLOCK TABLES;
```

```
--
-- Table structure for table `Worker`
--

DROP TABLE IF EXISTS `Worker`;
/*!40101 SET @saved_cs_client     = @@character_set_client */;
/*!40101 SET character_set_client = utf8 */;
CREATE TABLE `Worker` (
  `EmployeeID` int NOT NULL AUTO_INCREMENT,
  `Name` varchar(255) DEFAULT NULL,
  `BirthDate` date DEFAULT NULL,
  `Gender` enum('M','F') DEFAULT NULL,
  `Position` varchar(255) DEFAULT NULL,
  `WorkshopID` int DEFAULT NULL,
  PRIMARY KEY (`EmployeeID`),
  KEY `WorkshopID` (`WorkshopID`),
  KEY `idx_WorkerName` (`Name`),
  CONSTRAINT `worker_ibfk_1` FOREIGN KEY (`WorkshopID`) REFERENCES
`Workshop` (`WorkshopID`)
) ENGINE=InnoDB AUTO_INCREMENT=7 DEFAULT CHARSET=utf8mb4
COLLATE=utf8mb4_0900_ai_ci;
/*!40101 SET character_set_client = @saved_cs_client */;

--
-- Dumping data for table `Worker`
--

LOCK TABLES `Worker` WRITE;
/*!40000 ALTER TABLE `Worker` DISABLE KEYS */;
```

```
INSERT INTO `Worker` VALUES (1,'工人 1','1990-01-01','M','高级工程师',1),(2,'工人 2','1991-02-02','F','技术员',1),(3,'工人 3','1992-03-03','M','工程师',1),(4,'工人 4','1993-04-04','F','工程师',2),(5,'工人 5','1994-05-05','M','技术员',2);
/*!40000 ALTER TABLE `Worker` ENABLE KEYS */;
UNLOCK TABLES;

--
-- Table structure for table `Workshop`
--

DROP TABLE IF EXISTS `Workshop`;
/*!40101 SET @saved_cs_client     = @@character_set_client */;
/*!40101 SET character_set_client = utf8 */;
CREATE TABLE `Workshop` (
  `WorkshopID` int NOT NULL AUTO_INCREMENT,
  `DirectorName` varchar(255) DEFAULT NULL,
  `Address` varchar(255) DEFAULT NULL,
  `Phone` varchar(11) DEFAULT NULL,
  `FactoryName` varchar(255) DEFAULT NULL,
  PRIMARY KEY (`WorkshopID`),
  KEY `FactoryName` (`FactoryName`),
  CONSTRAINT `workshop_ibfk_1` FOREIGN KEY (`FactoryName`)
REFERENCES `Factory` (`FactoryName`)
) ENGINE=InnoDB AUTO_INCREMENT=3 DEFAULT CHARSET=utf8mb4
COLLATE=utf8mb4_0900_ai_ci;
/*!40101 SET character_set_client = @saved_cs_client */;

--
-- Dumping data for table `Workshop`
--
```

```
LOCK TABLES `Workshop` WRITE;
/*!40000 ALTER TABLE `Workshop` DISABLE KEYS */;
INSERT INTO `Workshop` VALUES (1,'李四','地址 1','12345678901','工厂 A'),(2,'
王五','地址 2','12345678902','工厂 A');
/*!40000 ALTER TABLE `Workshop` ENABLE KEYS */;
UNLOCK TABLES;

--
-- Final view structure for view `vw_productparts`
--

/*!50001 DROP VIEW IF EXISTS `vw_productparts`*/;
/*!50001 SET @saved_cs_client                = @@character_set_client */;
/*!50001 SET @saved_cs_results               = @@character_set_results */;
/*!50001 SET @saved_col_connection           = @@collation_connection */;
/*!50001 SET character_set_client            = utf8mb3 */;
/*!50001 SET character_set_results           = utf8mb3 */;
/*!50001 SET collation_connection            = utf8mb3_general_ci */;
/*!50001 CREATE ALGORITHM=UNDEFINED */
/*!50013 DEFINER=`root`@`localhost` SQL SECURITY DEFINER */
/*!50001 VIEW `vw_productparts` AS select `p`.`ProductID` AS
`ProductID`,`p`.`Price` AS `Price`,`pt`.`PartID` AS `PartID`,`pt`.`Price` AS
`PartPrice` from ((`product` `p` join `productpart` `pp` on((`p`.`ProductID` =
`pp`.`ProductID`))) join `part` `pt` on((`pp`.`PartID` = `pt`.`PartID`))) */;
/*!50001 SET character_set_client            = @saved_cs_client */;
/*!50001 SET character_set_results           = @saved_cs_results */;
/*!50001 SET collation_connection            = @saved_col_connection */;
/*!40103 SET TIME_ZONE=@OLD_TIME_ZONE */;
```

```
/*!40101 SET SQL_MODE=@OLD_SQL_MODE */;
/*!40014 SET FOREIGN_KEY_CHECKS=@OLD_FOREIGN_KEY_CHECKS */;
/*!40014 SET UNIQUE_CHECKS=@OLD_UNIQUE_CHECKS */;
/*!40101 SET
CHARACTER_SET_CLIENT=@OLD_CHARACTER_SET_CLIENT */;
/*!40101 SET
CHARACTER_SET_RESULTS=@OLD_CHARACTER_SET_RESULTS */;
/*!40101 SET
COLLATION_CONNECTION=@OLD_COLLATION_CONNECTION */;
/*!40111 SET SQL_NOTES=@OLD_SQL_NOTES */;

-- Dump completed on 2024-12-16    0:13:28
```